

Advancing Legal Reasoning in Algerian Law

RAG · KNOWLEDGE GRAPHS · ARGUMENT MINING → Citation-faithful legal answers

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01 The Problem

WHY WHY LEGAL QA IS HARD

1.1 Fluent is not lawful

A legal answer cannot be judged by fluency or plausibility. To be valid it must be **grounded in identifiable sources**, cite the **exact governing articles**, stay inside the **Algerian jurisdiction**, and read each provision **as amended** at the relevant date — and where the law is silent, the only correct answer is a **refusal**. A fluent language model guarantees none of this.

SOURCED · ARTICLE-CITED · JURISDICTION-BOUND · AMENDMENT-AWARE · OR REFUSED

1.2 What legal questions demand

بيّن الأساس القانوني لمبدأ خضوع القاضي للنص التشريعي، وحدود سلطته عند غياب النص...

— demands exact articles — Civil Code, arts. 1–2

كيف تغيّرت شروط الخلع في قانون الأسرة بعد تعديل 2005؟

— demands the law as amended — Ord. 05-02 → Family Code, art. 54

هل يحق لصاحب العمل في الجزائر فصل العامل؟
(at-will) أسباب تامة دون إبداء أسباب

— a foreign-doctrine trap — Algerian law is silent; the system must abstain

verbatim from AlgerianLegalBench (244 questions)

1.3 Where today's LLMs fail

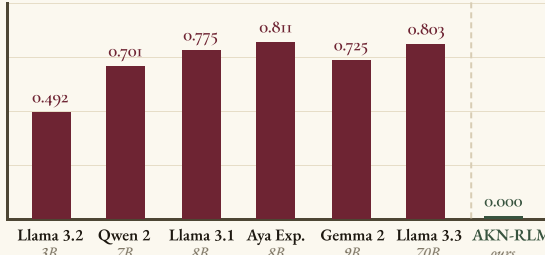


Fig. 1. Hallucinated-citation rate: the share of cited articles that *do not exist*, per direct-LLM baseline (244 questions). Scale does not help — the 70B model fabricates 4 of 5 citations. Gemini 2.5 reaches 0.008 only by declining to cite at all, and six of seven baselines import foreign law on >12 of 40 trap questions. AKN-RLM: 0.000.

1.4 ...and Algerian law raises the bar

- Arabic-first. 232 of 244 benchmark questions are in Arabic
- Layered in time. 1963–2025; amendments rewrite articles in place
- Deeply structured. code → chapter → section → article
- Low-resource. barely covered by existing legal-AI corpora

1.5 Objectives

- ✓ Structure the law
171 docs → Akoma Ntoso XML + legal KG
- ✓ Benchmark it
AlgerianLegalBench: 244 expert questions
- ✓ Reason on evidence
typed retrieval + bounded recursion
- ✓ Verify or abstain
triple faithfulness gate

1.6 Significance & impact

- First citation-faithful QA system for Algerian law — every claim resolves to a registered article, or the system abstains.
- Reusable infrastructure for Algerian legal AI
 - AKN corpus + knowledge graph — 171 structured documents, 1963–2025, article-level registry
 - AlgerianLegalBench — 244 expert-annotated questions, $\kappa = 0.829$
- A recipe for low-resource legal AI — evidence control beats parameter scale.

02 The Solution

HOW STRUCTURED CORPUS → BENCHMARK → AKN-RLM

I Structured legal corpus

the law, made machine-readable

What is Akoma Ntoso? The OASIS LegalDocML standard — an XML vocabulary for legal texts in which every law, article, amendment and cross-reference becomes an **addressable node** under a canonical FRBR URI: the article registry that every citation must resolve against.

WHAT WE BUILT

171 source files 189 MB 45 laws & codes → AKN XML 46 RDF graphs → one legal KG 1963–2025 coverage, amendments in place

```
<!-- Civil Code – Ordinance 75-58, as amended -->
<act name="code_civil">
  <FRBRWork value="/akn/dz/act/1975-09-26/75-58"/>
  <FRBRExpression value=".../ara@2005-06-20"/>
  <lifecycle>
    <eventRef date="2005-06-20" type="amendment"
      source="#law-05-10"/>
  </lifecycle>
  <article eId="art_124">
    <num>124</num>المادة</num>
    <content>كل...فعل أيًا كان يرتكبه الشخص بخطئه</content>
    <ref href="#art_127"/>
  </article>
</act>
```



Validated in **multiple manual passes** by two law students and a supervising senior legal expert — the structure is derived from the sources, never invented.

II AlgerianLegalBench

the evaluation layer

244 expert-annotated questions across 23 legal categories — 177 answerable, 67 expecting **abstention** · 232 Arabic / 12 French · each carries expected articles, a reference answer and a **four-step reasoning chain** · double-annotated, $\kappa = 0.829$.

QUERY TYPES

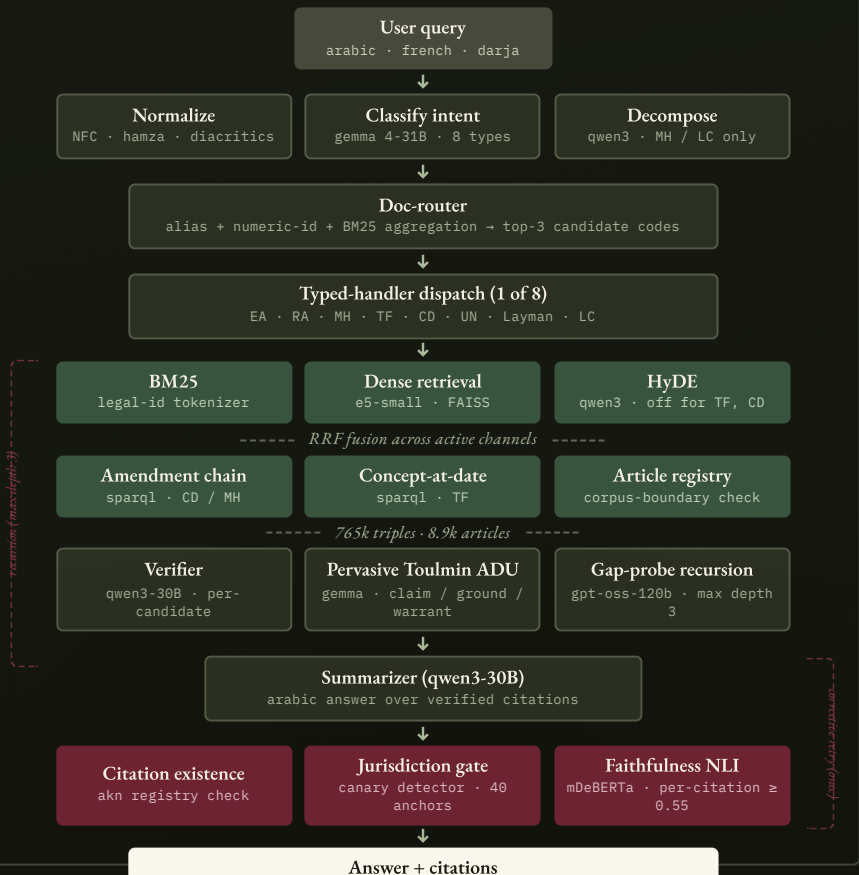
Rule application	66
Exact article	59
Unanswerable	40
Multi-hop	26
Layman	17
Long-context	17
Conceptual	12
Temporal	7

DIFFICULTY

23%	35%	42%
Easy	Medium	Hard
40 jurisdictional-infection traps — questions seeded with French, US, Egyptian, Tunisian & Gulf canary doctrine the system must refuse to import.		

III AKN-RLM

the answer pipeline — Fig. 3.2



03 Results

PROOF FROZEN 244-QUESTION RUN

0.305 Citation F1

article-level · locked

0.000

HCR · JIR

0.703

Abstention F1

1.64×

vs best direct LLM

1.74×

vs minimal RAG

2.9×

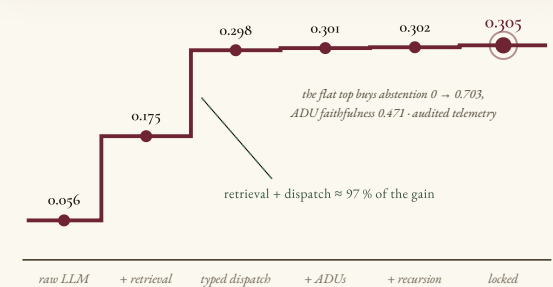
vs determ. retrieval

+0.199 over the strongest same-corpus baseline, $p_{Halm} < 0.0001$ · document-level F1 0.546

rlm_dispatched_full_phase_e_final · 2026-05-14 · depth ≤ 3 · NLI ≥ 0.55 · one retry → abstain

The ablation ladder

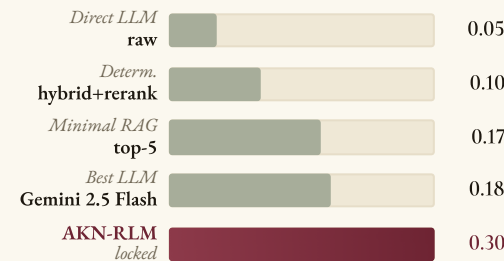
CITATION F1 BY PHASE



Each phase is an ablation in disguise — every layer pays for itself, though not always in F1.

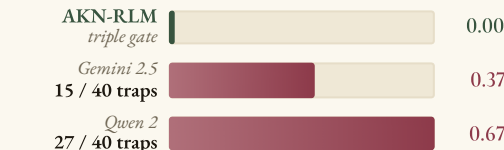
Citation F1 vs baselines

HIGHER = BETTER



Jurisdictional infection (JIR)

LOWER = SAFER



Only AKN-RLM holds 0.000 hallucinated-citation & foreign-law rates. Reliability tracks legal structure, not scale — the largest baselines hallucinate most.

Reading the results

INTERPRETATION

- The gain is architectural
identical corpus, registry & indices as every tier-2 baseline — the layers, not the scale, earn the F1
- Strictness has a price
the gate over-rejects layman queries; wider evidence pools dilute citation discipline
- Refusal works
38 / 40 infection traps correctly refused — every baseline invents an answer

LIMITS

- Retrieval ceiling: $R@10 = 0.258$
- French under-served
- Abstention precision 0.654
- Metrics are proxies

NEXT

- Corpus-tuned embedder
- French ↔ Arabic retrieval
- Graph-retrieval redesign
- Expert legal review



READ THE THESIS



i Structured sources

ii Controlled retrieval

iii Recursive reasoning

iv Citation verification



OUR CVS